

PRASANNA KAILASH WARAD

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SUMMARY

Data and AI Engineer with 1.5 years of production experience across ETL pipelines, ML models, and LLM systems. Cut reporting latency 28% on Spark/Airflow pipelines serving 19,000+ stores, and collapsed a 5-stage hiring process to under 2 minutes with an agentic platform.

TECHNICAL SKILLS

Data Engineering: ETL/ELT, Apache Spark, PySpark, Airflow, Data Modeling, Query Optimization, Data Quality Validation

LLM & Agentic AI: RAG, Hybrid Search (BM25 + Vector), Reciprocal Rank Fusion, MMR Retrieval, RAGAS Evaluation, LangChain, n8n, Groq

ML & Statistics: scikit-learn, PyTorch, Pandas, NumPy, Classification, Model Evaluation (F1, ROC-AUC), DeepFace, MediaPipe

Databases & Cloud: PostgreSQL, pgvector, ChromaDB, Pinecone, Supabase, Snowflake, AWS (S3, Redshift, CloudWatch)

Languages & Backend: Python, SQL (Advanced), TypeScript, R, Bash, FastAPI, Flask, Node.js, Express, Docker, GitHub Actions

EXPERIENCE

Cloud BC Labs

Reston, VA

Software Engineer Intern, AI & Data Engineering

Oct 2025 – May 2026

- Automated the candidate assessment lifecycle with an n8n agentic workflow and Groq LLaMA-3.3-70b: job-description submission triggers question generation and email delivery, cutting a 5-stage process to under 2 minutes.
- Orchestrated 4 AI services into a 6-container Docker Compose pipeline with Ollama fallback: DeepFace identity verification, Groq MCQ scoring, Whisper transcription, and MediaPipe proctoring.
- Designed 6 PostgreSQL schema migrations from scratch, normalizing candidates, assessments, AI scores, and proctoring flags into a schema powering real-time recruiter dashboards.
- Built FastAPI, Flask, and Node.js/Express REST APIs with JWT auth linking the React dashboard to AI scoring services, standardizing contracts across 4 microservices and cutting integration overhead 20%.

HCLTech Ltd

Noida, India

Data Engineer, Dollar General (Fortune 100 Retailer)

Feb 2023 – Jul 2024

- Optimized Apache Spark and Airflow ETL pipelines with partition pruning and query caching, cutting reporting latency 28% across a retail analytics stack serving 19,000+ stores.
- Validated 2M+ retail transaction records with Python and SQL anomaly detection, resolving data quality gaps degrading inventory replenishment accuracy.
- Built an anomaly detection and downtime forecasting model with scikit-learn and PyTorch on AWS CloudWatch logs, cutting incident response 25%.

PROJECTS

SEC Filing Intelligence: RAG over SEC EDGAR | *LangChain, ChromaDB/Pinecone, Groq, FastAPI* | [GitHub](#)

- Built an end-to-end RAG system over SEC 10-K/10-Q filings returning source-cited answers via LangChain LCEL, local BGE embeddings, and MMR retrieval with metadata filtering, backed by ChromaDB in dev and Pinecone in prod.
- Engineered LLM cost guardrails (token budgets, rate throttling, caching, retrieval-only degradation) and validated quality with a RAGAS pipeline scoring faithfulness, answer relevancy, and context recall over a golden dataset in CI.

RAGBase: Production RAG Platform | *Next.js, TypeScript, Supabase/pgvector, Groq, Gemini* | [GitHub](#)

- Built a document-intelligence system with hybrid retrieval fusing BM25 keyword scoring and vector similarity via Reciprocal Rank Fusion, plus token-by-token streaming with source-attributed citations.
- Engineered a four-layer provider fallback chain (Groq primary, Gemini backup) across embeddings and chat generation with pgvector persistence, deployed on Vercel with Supabase Postgres.

CardioRisk: Predictive Health Analytics Pipeline | *R, Statistical Modeling, Classification*

- Benchmarked 5 classification models on 319K+ CDC health records in R; Naive Bayes led at 91% accuracy and 0.81 AUC, and flagged a Decision Tree matching it at 0.50 AUC as majority-class prediction, not signal.
- Ranked BMI, age category, sleep duration, and general health as top predictors across 17 clinical attributes via Random Forest Gini importance.

EDUCATION

The University of Texas at Dallas

Richardson, TX

M.S., Information Technology & Management, Dean's Excellence Scholar | *GPA: 3.88/4.0* | *SnowPro Core*

Aug 2024 – May 2026

Savitribai Phule Pune University

Pune, India

B.E., Electrical Engineering | *GPA: 3.80/4.0*

Aug 2018 – May 2022